

Radiation Hydrodynamics of Self-gravitating Protoplanetary Disks. II. Accretion, Migration, Spin-up, Contraction and Final Fates of GI Fragments

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ABSTRACT

In the first paper of this series we showed that gravitational instability (GI) in protoplanetary disks can produce fragments of planetary rather than brown-dwarf mass, leaving open what such fragments become. Here we follow every bound object in a global three-dimensional radiation hydrodynamic simulation of a fragmenting $0.196 M_{\odot}$ disk around a $1 M_{\odot}$ star: seven surviving fragments, one disrupted clump and one merged clump, followed for 1.2 kyr with per-object mass and angular-momentum budgets. The fragments form at $1.4\text{--}3.0 M_{\text{J}}$, consistent with the normalized initial-mass distribution of Paper I. Growth follows a single Hill-limited scaling $\propto \Sigma \Omega R_{\text{H}}^2$, regulated by the delivery of gas into draining feeding zones, and the latest-forming fragments starve. Migration is bidirectional and controlled by the gravitational interactions with not only the disk but also neighboring clumps. Accretion sets most spins, which align with the orbits and supply a tenth of the support against gravity. The interiors are entropy-stratified and largely convectively stable, and grow far faster than they can cool: compression is quasi-adiabatic, and central entropies near $12 k_{\text{B}}$ per baryon favor hot-start initial conditions for those that end as gas giants. Integrating the measured growth law against the measured gas budget, we predict that our disk delivers gas giants, brown dwarfs and a possible low-mass stellar companion, while one scattered fragment may add a free-floating planet: the outcome of disk fragmentation is set not only at birth but by the mass supply and dynamics that follow.

Keywords: Gravitational instability (668) — Protoplanetary disks (1300) — Planet formation (1241)
— Radiative transfer (1335)

1. INTRODUCTION

Gravitational instability (GI) is one of the two principal pathways proposed for giant planet formation (A. G. W. Cameron 1978; A. P. Boss 1997; K. Kratter & G. Lodato 2016), complementary to core accretion (J. B. Pollack et al. 1996; M. Lambrechts & A. Johansen 2012) at the wide separations and low stellar masses where the latter struggles. The long-standing objection to it is one of scale: disk fragmentation was generally expected to produce over-massive clumps, closer to brown dwarfs than to gas giants (D. Forgan & K. Rice 2011; K. Kratter & G. Lodato 2016; W. Xu et al. 2025). In the first paper of this series (Y. Ni et al. 2025a, hereafter Paper I) we addressed this objection with global radiation hydrodynamic (RHD) simulations that resolve the local collapse: fragments are born at planetary masses, dis-

tributed around a universal mass scale set by the local conditions at the fragmenting spiral arms. Birth mass, however, is only the beginning.

What a fragment becomes depends heavily on everything that follows its formation: how fast it accretes and for how long, where it migrates, how much angular momentum it acquires, and how its interior responds to the mass it gains. The theoretical picture of the above physical evolution of fragments is still incomplete.

The fragment accretion rate is set by how much of the surrounding disk a fragment can capture, and the shear-limited estimates differ by an order of magnitude in normalization. A. C. Boley et al. (2010) capped delivery into the Hill sphere at $\dot{m} \simeq 0.5 \Sigma \Omega R_{\text{H}}^2$, while Z. Zhu et al. (2012) calibrated a capture radius of $1.7 R_{\text{H}}$ against two-dimensional simulations and obtained $\dot{m} \simeq 4 \Sigma \Omega R_{\text{H}}^2$, eight times larger. A three-dimensional radiation hydrodynamic calculation of a collapsing cloud core in-

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Instead, we found Bondi-limited growth at $10^{-5} M_{\odot} \text{yr}^{-1}$ (Y.

(Tsukamoto et al. 2013). The distinction matters because the asymptotic mass is large. Fragments that keep accreting reach isolation masses of order $0.1 M_{\odot}$ (Z. Zhu et al. 2012), and K. M. Kratter et al. (2010) argued from the same reasoning that disk-born objects should be brown dwarfs and low-mass stars, with planets only their low-mass tail. The tidal-downsizing scenario resolves this instead by removing mass, the fragment migrating inward until the star strips its envelope and leaves a solid core if grains had time to sediment first (S. Nayakshin 2017).

The migration can be fast but its direction is disputed. Planets inserted into gravito-turbulent disks migrate inward on the linear Type-I timescale, about ten orbits at 100 au, with no time to open a gap (C. Baruteau et al. 2011), and clumps that form self-consistently in a two-dimensional irradiated disk begin the same way, four of thirteen being tidally destroyed on the way in and three growing massive enough to open a gap and stall (Z. Zhu et al. 2012), as an accreting protoplanet also does (D. Stamatellos 2015). Other work finds the opposite. Gravitationally unstable gap edges drive outward dynamical migration (R. Cloutier & M.-K. Lin 2013), an embedded planet reverses direction as it couples to spiral modes (S. Michael et al. 2011), and inserted planets slow in the gravitationally stable inner disk without opening a gap (S. Rowther & F. Meru 2020). Where several fragments coexist they perturb one another (E. I. Vorobyov & V. G. Elbakyan 2018), and the most recent three-dimensional simulations find migration that is stochastic in both direction and rate, driven by a local torque from within about two Hill radii (N. Kubli et al. 2026).

The specific angular momentum a fragment inherits from the shear of its parent arm exceeds that of Jupiter’s present spin by some three orders of magnitude (A. C. Boley et al. 2010), so most of it has to go somewhere. Three-dimensional simulations find protoplanets rotating differentially and staying nearly spherical (L. Mayer et al. 2004), with rotation supplying about a fifth of their support against gravity when they begin to collapse, after which an $m = 2$ mode carries the angular momentum outward (M. Galvagni et al. 2012). Clumps formed in magnetized disks rotate more slowly still (N. Kubli et al. 2023), and encounters between fragments can leave a clump spinning retrograde to its own orbit (C. Hall et al. 2017). What the shed angular momentum builds is a circumplanetary disk, and simulations of the two channels under matched conditions predict that the one built by fragmentation is colder by more than an order of magnitude, which has been proposed as an observational discriminant (J. Szulágyi et al. 2017).

Fragment contraction calculations usually follow a single clump on its own, after it is separated from the disk that fed it (e.g., A. C. Boley et al. 2010; M. Galvagni et al. 2012). In one-dimensional models of this kind the time to H_2 dissociation scales with the assumed metallicity and falls to about a thousand years once grain growth and settling lower the opacity. Rotation and ongoing accretion, both of which could lengthen it, are left out (R. Helled & P. Bodenheimer 2011). No global three-dimensional RHD simulation has yet measured accretion, migration, spin-up and contraction acting together on a population of fragments embedded in their parent disk.

Population syntheses approach the same question from the other end, following whole populations of fragments to compare with surveys (e.g., D. Forgan & K. Rice 2013; S. Nayakshin 2017; D. H. Forgan et al. 2018; S. Müller et al. 2018; O. Schib et al. 2025a,b). Because the physics above is unsettled, they must supply accretion, migration and contraction as prescriptions, and the choices decide the answer. Whether fragments migrate inward or are stranded at wide separations moves the formation frequency inferred from direct imaging by an order of magnitude (J. Humphries et al. 2019). When one migrating-clump problem was put to seven hydrodynamic codes and to the prescriptions of earlier syntheses, the prescriptions disagreed among themselves far more than the codes did (M. Fletcher et al. 2019). Robust population predictions therefore wait on the first-principles calculations.

Observations call for the same measurements. Directly imaged companions at tens to hundreds of au (A. J. Bohn et al. 2020; A. Vigan et al. 2021) and giant planets around low-mass stars (J. C. Morales et al. 2019) are natural GI candidates, and the properties that could discriminate their origin are exactly the ones set during the embedded phase. The luminosity of a young object reflects the entropy its formation channel left behind (M. S. Marley et al. 2007; D. S. Spiegel & A. Burrows 2012), its spin and obliquity record its accretion and dynamical history (M. L. Bryan et al. 2018, 2021), and the circumplanetary disk that should carry the shed angular momentum has now been detected around a young companion (M. Benisty et al. 2021). Free-floating planets are a further candidate by-product of the same dynamics (N. Miret-Roig et al. 2022). Reading any of these back to a formation channel requires knowing how a fragment accretes, migrates and contracts while it is still embedded.

This paper makes those measurements in a single deep simulation. We follow every bound object in a fragmenting $0.196 M_{\odot}$ disk—seven surviving fragments, one

disrupted clump and one merged clump—for ≈ 1.2 kyr after the onset of fragmentation, with radiation transport, tabulated opacities and a molecular equation of state, and we close mass and angular-momentum budgets for each object. We first ask whether the initial masses reproduce the Paper I distribution under a different equation of state, opacity and disk model, and then what sets each fragment’s accretion rate, what drives its migration, where its spin comes from, how its interior evolves, and what the measured growth law and gas budget imply for its final mass.

The paper is organized as follows. Section 2 describes the simulation and the analysis conventions. Section 3 presents the fragment census, initial masses, accretion, migration, spin, interior structure, and conditional final masses. Section 4 compares with Paper I and with models of fragment evolution, discusses the observational signatures of an ejected fragment, and states the caveats. Section 5 summarizes.

2. METHODS

2.1. Numerical method

Our simulation is performed with the meshless finite-mass method in the GIZMO code (P. F. Hopkins 2015), coupled to the radiation-hydrodynamics module that we developed and tested in Paper I. The module solves the frequency-integrated radiation moment equations under the M1 closure (C. D. Levermore 1984) with a second-order Harten–Lax–van Leer transport solver with piecewise-linear reconstruction, an implicit–explicit treatment of the stiff radiation–matter coupling terms, and a geometric outflow boundary condition that lets thermal radiation escape from the disk surface. The governing equations, solver design, and an extensive test suite are described in Paper I, and we do not repeat them here.

The present run differs from the simulation suite of Paper I in the disk model and run setup of Section 2.2. First, the gas equation of state uses an adiabatic index $\gamma = 1.4$ with a mean molecular weight $\mu = 2.33$, appropriate for the cold, molecular (H_2 -dominated) gas of the outer disk, instead of the monatomic $\gamma = 5/3$ adopted in Paper I for comparability with earlier work. Rotational degrees of freedom of H_2 are therefore included in the internal energy, while H_2 dissociation is not modeled. We verify in Section 3.6 that even the fragment centers, the densest regions we track, remain below $\sim 10^3$ K and therefore safely outside the dissociation regime. Second, the gray power-law opacity of Paper I is replaced by the tabulated opacities of Z. Zhu et al. (2021), which pair solar-abundance gas with a dust mixture of size-distribution slope $q = 3.5$ and maximum grain size

$a_{\text{max}} = 0.1$ cm. The table provides distinct Rosseland and Planck means over $10^{-14} \leq \rho/(\text{g cm}^{-3}) \leq 0.79$ and $1 \leq T/\text{K} \leq 7.9 \times 10^6$. Third, the reduced speed of light is $\tilde{c}/c = 5 \times 10^{-4}$. It satisfies the static-diffusion criterion, and it lies within the range that Paper I demonstrated to be converged by quadrupling their fiducial $\tilde{c}/c = 4 \times 10^{-4}$ with little change (their Appendix B). Our cell mass ($1.0 \times 10^{-4} M_{\text{J}}$) is at the resolution level of the Paper I suite (Table 1), the level at which their convergence tests were performed.

2.2. Disk model

We simulate a single massive disk of $M_{\text{d}} = 0.196 M_{\odot}$ around a $1 M_{\odot}$ star, initially extending from approximately 60 to 180 au with a surface density profile $\Sigma \propto R^{-7/4}$ and a temperature profile $T \propto R^{-1/2}$, normalized to 75 K at 100 au. These two slopes place the disk at a uniform Toomre parameter $Q = c_s \Omega / (\pi G \Sigma) = 2.0$ across its full radial extent, where $c_s = (k_{\text{B}} T / \mu m_{\text{H}})^{1/2}$ is the isothermal sound speed and Ω the Keplerian angular frequency. The disk is resolved by 2,038,732 equal-mass cells of $9.63 \times 10^{-8} M_{\odot}$ ($1.0 \times 10^{-4} M_{\text{J}}$) each. The star is treated as an external point-mass potential. From this configuration the disk cools, develops rings and spiral arms, and forms its first bound clump at $t = 2.46$ kyr. We run for a total of 3.67 kyr (hereafter t_{end}), i.e., ≈ 3.7 orbits at 100 au, so the fragment population is followed for ≈ 1.2 kyr after the onset of fragmentation. Gas that spreads beyond the radial boundaries of the computational domain, at 30 and 300 au, is removed. Over the run this removes 1.7% of the disk ($3.6 M_{\text{J}}$), just over half of it through the inner boundary, the rest through the outer, and all of it far from the fragments’ orbits. Snapshots are output every 24.15 yr, and all Lagrangian analyses below exploit the unique cell identities across snapshots. Table 1 summarizes the setup alongside that of Paper I.

2.3. Clump identification, tracking, and the fragment sample

We identify bound structures in every snapshot with CloudPhinder⁴, the structure finder used in Paper I: starting from local density peaks, it grows candidate structures cell by cell in order of decreasing density and retains the largest structure around each peak that is energetically bound,

$$E_{\text{K}} + U < |E_{\text{P}}|, \quad (1)$$

where E_{K} , U , and E_{P} are the bulk kinetic energy (measured about the structure’s center-of-mass velocity), the

⁴ <https://github.com/mikegrudic/CloudPhinder>

Table 1. Simulation setup of this work compared with the suite of Paper I

Parameter	Paper I	This work
Star mass ($M_\odot$)	1	1
Disk mass ($M_\odot$)	0.13–0.20	0.196
Radial extent (au)	30–100	60–180
Σ profile	$\propto R^{-1}$	$\propto R^{-7/4}$
T profile	$\propto R^{-1/2}$	$\propto R^{-1/2}$
Initial Toomre Q^a	1.5 at 100 au	2.0 (uniform)
Adiabatic index γ	5/3	1.4
Mean molecular weight μ	...	2.33
Opacity	$\min[(T/100\text{ K})^2, 1] \kappa_0, \kappa_0 = 1 \text{ or } 5 \text{ cm}^2 \text{ g}^{-1}$	tabulated ^b
$\tilde{c}/c$	4×10^{-4}	5×10^{-4}
Number of cells	$\sim 2 \times 10^6$	2.04×10^6
Cell mass (M_J)	$\sim 10^{-4}$	1.0×10^{-4}
Output cadence (yr)	10	24.15
Duration (kyr)	≤ 5	3.67

^aSound-speed conventions differ with the equations of state. This work evaluates Q with the isothermal sound speed (Section 2.2).

^bRosseland and Planck mean opacities from Z. Zhu et al. (2021), described in Section 2.1.

internal energy, and the gravitational self-energy of the structure. Condition (1) states that the total energy is negative and is algebraically identical to the $\alpha_{\text{crit}} = 2$ selection of Paper I. We apply the finder to cells above a density threshold of $5.9 \times 10^{-14} \text{ g cm}^{-3}$ and retain structures of more than three cells. A *clump* is a structure satisfying Equation (1). For the census we additionally require that the structure persist over at least two consecutive snapshots, which excludes one marginally bound over-density that appeared and dispersed within a single output interval.

To measure how far a clump has proceeded toward virial equilibrium we use the virial parameter

$$\alpha_{\text{vir}} = \frac{2E_K + 3(\gamma - 1)U}{|E_P|}. \quad (2)$$

This definition differs deliberately from Equation (35) of Paper I. For a self-gravitating ideal gas the scalar virial theorem reads $2E_K + 3(\gamma - 1)U = |E_P|$, since the pressure integral is $3 \int P dV = 3(\gamma - 1)U$, and a structure in virial equilibrium therefore has $\alpha_{\text{vir}} = 1$ in Equation (2) for any value of γ . For the monatomic gas of Paper I, $3(\gamma - 1) = 2$ and Equation (2) reduces exactly to their $\alpha_{\text{vir}} = 2(E_K + U)/|E_P|$, so the two papers' virial parameters are directly comparable.

Clumps are linked across snapshots through the overlap of their member-cell identities (as in Y. Ni et al. 2025b), which yields a set of branches with unambigu-

ous formation epochs, one disruption, and one merger (Section 3.1). Following Paper I we define the *identification epoch* t_{identify} of a clump as the first snapshot at which it satisfies Equation (1). Its *initial mass* m_{frag} is its bound mass at that epoch and its initial orbital radius R_{frag} is the cylindrical radius of its center of mass at the same epoch. We measure its size by the effective radius $R_{\text{eff}} = [(5/3) \sum_i m_i |\mathbf{r}_i - \mathbf{r}_c|^2 / \sum_i m_i]^{1/2}$, with m_i and $\mathbf{r}_i$ the mass and position of member cell i and $\mathbf{r}_c$ their center of mass, as in Paper I.

We define *fragments* as the clumps that survive, without disruption, to the end of the run. A clump whose branch terminates by merging into a surviving clump is counted as part of the survivor from the merger epoch onward, and the merger is recorded as a discrete event in the survivor's history. Seven fragments result, which we label F1–F7 in order of formation, F6 and F7 forming at the same snapshot with different initial masses. The disrupted clump is labeled D1, and the late-forming clump that merges into F1 is labeled M1. Every survivor reaches $\min[\alpha_{\text{vir}}(t)] = 0.99\text{--}1.16$ during the run, i.e., all seven approach virial equilibrium in the same sense in which the fragments of Paper I approach $\alpha_{\text{vir}} \rightarrow 1$. We therefore do not impose a separate virial threshold in the definition.

2.4. Analysis conventions

We define several quantities used in Section 3 here. The definitions come in seven groups: ambient gas and local disk quantities, mass accretion, growth episodes, torques, spin, interior profiles, and the matched-age comparison.

Ambient gas and local disk quantities. The *ambient* gas at a snapshot is all gas belonging to no cataloged clump. Local disk quantities around a fragment are computed from ambient gas only, so that a fragment’s own envelope and its neighbors’ envelopes do not contaminate its environment. A fragment’s orbital radius a is the cylindrical radius of its center of mass, and orbital frequencies are taken Keplerian, $\Omega = (GM_\star/a^3)^{1/2}$, as an ad hoc approximation. We note that the measured rotation of the ambient gas deviates from the Keplerian rate by several percent, and locally more within arms, through the disk’s self-gravity and pressure gradients, but the quantities built from Ω , namely the rate scale $\Sigma_{\text{loc}}\Omega R_{\text{H}}^2$, the scale height H , the linear migration timescale, and the analytic initial-mass scale of Section 3.2, are used either with measured normalizations or in comparisons at the factor-of-a-few level, so the deviation shifts no conclusion. The orbital eccentricity e is the two-body osculating value computed from the center-of-mass position $\mathbf{r}$ and velocity $\mathbf{v}$ about the star alone,

$$e = \left| \frac{\mathbf{v} \times (\mathbf{r} \times \mathbf{v})}{GM_\star} - \frac{\mathbf{r}}{|\mathbf{r}|} \right|. \quad (3)$$

The Hill radius is $R_{\text{H}} = a[m/(3M_\star)]^{1/3}$, with m the fragment’s bound mass. The local surface density Σ_{loc} is the ambient mass within the in-plane circle of radius $2R_{\text{H}}$ around the fragment, divided by the area of that circle. The local sound speed c_s is the mass-weighted isothermal sound speed of the same gas, the local scale height is $H = c_s/\Omega$, and the thermal (Bondi) radius is $R_{\text{B}} = Gm/c_s^2$. The *feeding annulus* of a fragment is the region of the disk between $a - 2R_{\text{H}}$ and $a + 2R_{\text{H}}$, and its ambient mass is M_{feed} . Section 3.3 shows that this extent, $2R_{\text{H}}$ on each side of the orbit, matches the measured origins of the gas the fed fragments F1–F4 accrete.

Mass accretion. Two quantities enter the mass budget: the Hill inventory $M(< R_{\text{H}})$, which is the total gas mass inside the sphere of radius R_{H} centered on the fragment, and its unbound part $M_{\text{res}} = M(< R_{\text{H}}) - m$. The *total inflow* over a stated interval is the mass that crosses the Hill surface inward during it, counted without subtracting the mass that crosses outward. The growth rate $\dot{m}$ used throughout is the net rate of change of the bound mass, smoothed with a centered running mean of width equal to the orbital shear time Ω^{-1} , eval-

uated at each fragment’s median orbital radius. Growth rates are compared with the Hill rate through the efficiency $C_{\text{eff}} \equiv \dot{m}/(\Sigma_{\text{loc}}\Omega R_{\text{H}}^2)$, whose denominator is smoothed over the same window as $\dot{m}$. We discard intervals in which more than 5% (mass) of the gas in a fragment’s Hill sphere lies simultaneously inside a neighboring clump’s Hill sphere and, for F1, the intervals spanning the merger. For the efficiency, discarded intervals are masked in both numerator and denominator before the smoothing, and efficiency statistics use the intervals with positive smoothed rate. We quote the median of C_{eff} for each fragment alone and the median over the pooled intervals of all seven.

Growth episodes. Alongside the smooth growth we identify discrete dynamical events, short intervals in which the fragment mass changes much faster than smooth accretion alone would produce. Section 3.3 classifies the events by mechanism.

Torques. The gravitational torque on a fragment is measured from the cell-scale acceleration from the simulation output: with $\mathbf{g}_i$ the total gravitational acceleration on member cell i ,

$$\Gamma_{\text{ext}} = \sum_{i \in \text{members}} m_i (\mathbf{r}_i \times \mathbf{g}_i)_z, \quad (4)$$

in which the star, whose force on every member is central, and the members’ mutual self-gravity, which cancels pairwise, contribute exactly zero. The sum is therefore the torque exerted by all matter outside the fragment. We split it exactly by calculating the gravitational acceleration of every member cell from two separate source sets: the member cells of every other cataloged clump, which gives Γ_{clumps} , and the ambient gas, which gives Γ_{gas} . The hydrodynamic torque $\sum m_i (\mathbf{r}_i \times \mathbf{a}_{\text{hyd},i})_z$, evaluated from the cells’ hydrodynamic accelerations, enters the angular-momentum budget of Section 3.4. The radial drift rate $\dot{a}$ is the change of a between consecutive snapshots divided by the output interval. The drift timescale is the per-interval $|a/\dot{a}|$, and comparisons with the linear timescale quote the median of the per-interval ratio over each fragment’s life. The linear (Type-I) reference timescale is $t_{\text{I}} = (1/2.7)(M_\star/m)(M_\star/\Sigma a^2)(H/a)^2\Omega^{-1}$ (the constant- Σ coefficient of H. Tanaka et al. 2002), evaluated from the local ambient disk, with Σ taken from the feeding annulus and H as above.

Spin. The spin of a fragment is the angular momentum of its bound members about their center of mass, in the center-of-mass velocity frame,

$$\mathbf{S} = \sum_{i \in \text{members}} m_i (\mathbf{r}_i - \mathbf{r}_c) \times (\mathbf{v}_i - \mathbf{v}_c). \quad (5)$$

Here $\mathbf{r}_c$ and $\mathbf{v}_c$ are the mass-weighted mean position and velocity of the members. To separate ordered rotation from random motions, we divide the members into twelve cylindrical shells of equal cell number about the spin axis $\hat{\mathbf{S}} = \mathbf{S}/|\mathbf{S}|$ and define the rotational energy as the kinetic energy of the mean azimuthal motion of those shells,

$$E_{\text{rot}} = \sum_k \frac{1}{2} M_k \langle v_\phi \rangle_k^2, \quad (6)$$

where M_k is the mass of shell k and $\langle v_\phi \rangle_k$ the mass-weighted mean azimuthal velocity about $\hat{\mathbf{S}}$ within it. The rotational support is then $\beta_{\text{spin}} = E_{\text{rot}}/|E_P|$. The spin-orbit obliquity is the angle between $\mathbf{S}$ and the orbital angular momentum $m\mathbf{r}_c \times \mathbf{v}_c$. The degree of differential rotation is the logarithmic slope $q_\Omega = d \ln \Omega_{\text{spin}} / d \ln \varpi$ of the shell rotation profile $\Omega_{\text{spin}}(\varpi) = \langle v_\phi \rangle(\varpi)/\varpi$, with ϖ the cylindrical distance from the spin axis. The slope is fitted by least squares in the log-log plane over the shells whose mean radii lie between the radii enclosing one quarter and one half of the bound mass. The spin delivered by accretion, ΔJ_{adv} , is the component of the spin angular momentum along the fixed rotation axis of the disk carried by joining cells minus that carried by leaving cells, evaluated in each epoch's center-of-mass frame. We report its running total as a fraction of the change of the spin component along the same fixed axis, ΔJ_{spin} , accumulated to the same time.

Interior profiles. Entropy is traced by the adiabat label

$$K \equiv (\gamma - 1) u \rho^{1-\gamma}, \quad (7)$$

where u is the specific internal energy of a cell, so that $U = \sum_i m_i u_i$ over the bound members. The label is proportional to P/ρ^γ and is therefore constant for any parcel of ideal gas that evolves adiabatically. A rise in K measures heating beyond what compression alone provides, and a fall measures cooling.

Because the gas obeys a fixed- γ equation of state, K also fixes its specific entropy. With $c_V = k_B/(\gamma - 1)$ per particle, the entropy per baryon is

$$s = \frac{k_B}{\mu(\gamma - 1)} \ln \frac{K}{K_0} + s_0, \quad (8)$$

with $1/[\mu(\gamma - 1)] = 1.073$ for our gas, so that only the constant s_0 is undetermined. We fix it on the ideal-gas expression of G.-D. Marleau & A. Cumming (2014), which gives $s_0 = 9.6 k_B$ per baryon at $K_0 = 2.9 \times 10^{12} \text{ erg g}^{-1} (\text{g cm}^{-3})^{-2/5}$, the value of K at 1600 K and 3 bar. Our entropies then carry the zero point that G.-D. Marleau & A. Cumming (2014) use for the initial

entropies of young giant planets. In the following analysis, we quote the central value, defined with the other central quantities below.

Interior profiles are constructed at every output snapshot of a fragment's life, about its density peak, defined as the density-weighted centroid of its densest 1% of cells. Profiles, in radius or in mass coordinate, are medians in 32 shells of equal cell number, which is equal mass since the cells have equal masses. In interior contexts the half-mass radius R_{half} is the median member radius about that density peak. Central values are medians over the same densest 1%. Convective stability is judged by the Schwarzschild criterion for an adiabatic stratification, $dK/dr \geq 0$. The free-fall time is $t_{\text{ff}} = [3\pi/(32G\rho_c)]^{1/2}$ at the central density ρ_c , the accretion time is $t_{\text{acc}} = m/\langle \dot{m} \rangle$ with $\langle \dot{m} \rangle$ the lifetime-mean growth rate, and $L = 4\pi R_{\text{half}}^2 \langle F_r \rangle$ is the luminosity obtained from the mean outward radiative flux $\langle F_r \rangle$ of the cells whose radii lie within ten percent of the half-mass radius. The Kelvin-Helmholtz time is $t_{\text{KH}} = |E_P|/L$, the time in which a fragment could radiate its binding energy away at that luminosity. The accretion power is $L_{\text{acc}} = Gm\langle \dot{m} \rangle/R_{\text{half}}$.

Matched age. Where fragments are compared at a common age we use the first 434.7 yr after each fragment's identification (18 output intervals). This is the age of the youngest fragments at the end of the run, and therefore the longest span over which all seven can be compared.

3. RESULTS

3.1. Fragment census and fates

Figure 1 shows the disk at the onset of fragmentation and at the end of the run. The first bound clump, F1, emerges at $t = 2.46$ kyr at an orbital radius of 122 au, and eight more bound clumps follow over the next kiloyear, the last of them M1 at 3.41 kyr (Table 2). M1 survives as a separate clump for only 0.12 kyr before merging into F1 at 3.53 kyr. By the end of the run the disk is crowded: seven fragments of 1.7–20 M_J orbit between 79 and 169 au, and their gravitational interplay, rather than coherent spiral structure, dominates the morphology.

Figure 2 presents the census. Of the nine distinct bound clumps, seven survive to the end of the run (F1–F7, in order of formation), and their masses, identification epochs, and fates are collected in Table 2. The two failures illustrate two distinct channels. D1, born between F2 and F3, is disrupted after 0.2 kyr. M1 migrates inward until its merger with F1, and we treat its mass as joining F1 from that epoch. The virial parameter separates the outcomes cleanly (Figure 2b): ev-

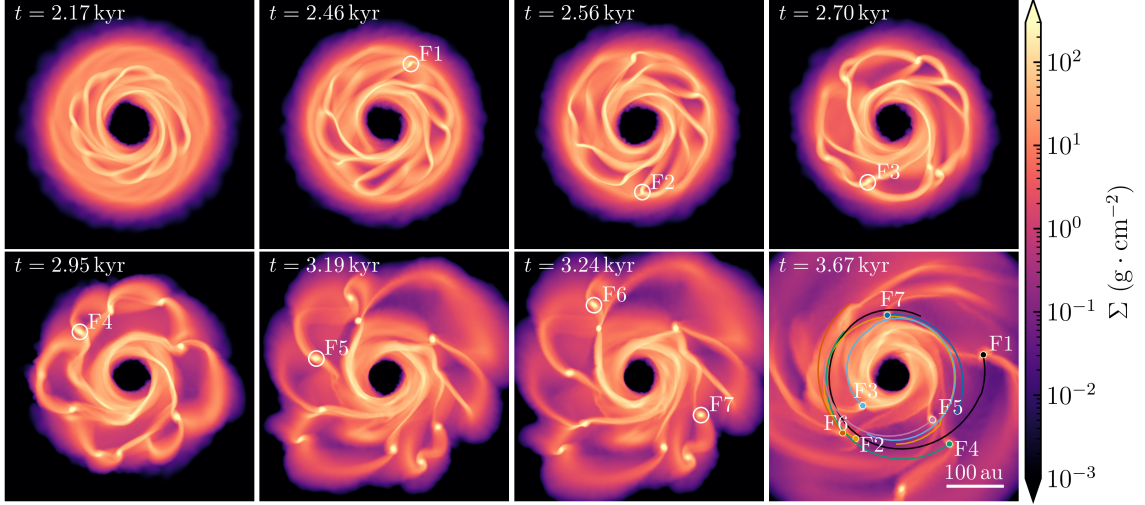


Figure 1. Gas surface density before fragmentation ($t = 2.17$ kyr, top left), at the identification epoch of each fragment (each newly identified fragment circled and labeled, and F6 and F7 sharing the $t = 3.24$ kyr panel), and at the end of the run (bottom right), where the trajectories of the seven fragments from birth to the end of the run are overlaid (colored curves, labels at final positions). Note the transition from a smooth ringed and spiral disk to the crowded, interaction-dominated late-time state.

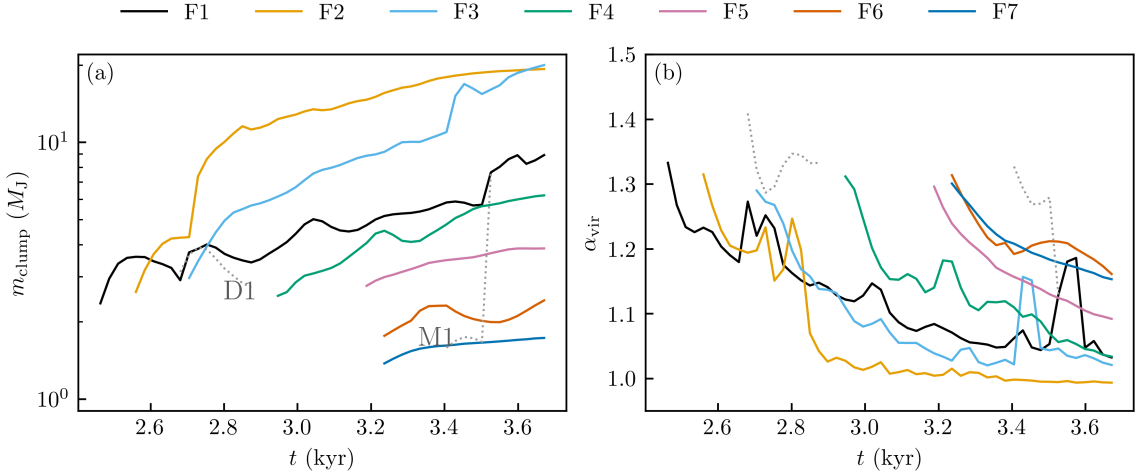


Figure 2. Bound-clump census. (a) Mass evolution of all tracked clumps: the seven fragments F1–F7 (colored solid curves) and the disrupted and merging clumps D1 and M1 (gray dotted). M1’s curve joins F1’s at the merger. (b) The virial parameter of Equation (2). Fragments descend toward $\alpha_{\text{vir}} = 1$ (minima 0.99–1.16) while D1 and M1 turn around at $\alpha_{\text{vir}} \gtrsim 1.27$ before disruption or merger.

every survivor descends toward $\alpha_{\text{vir}} = 1$, reaching minima of 0.99–1.16, whereas D1 and M1 turn around at $\alpha_{\text{vir}} \gtrsim 1.27$ before their disruption or merger. This is the same separation between collapsing fragments and dissolving clumps found in Paper I.

3.2. Initial fragment masses

Figure 3a shows the initial masses of the seven fragments: $m_{\text{frag}} = 1.4\text{--}3.0 M_J$. The failed objects are not distinguishable by initial mass, since M1 ($1.6 M_J$) lies within the range of the fragments and D1 ($3.1 M_J$)

lies only marginally above its upper end. Survival is therefore decided by the dynamics that follow formation rather than by the initial collapse.

Paper I found that initial fragment masses follow the non-axisymmetric spiral-arm collapse estimate $M_{\text{analytical}} = \Sigma \cdot \lambda_T \cdot 2c_s/\Omega$, where $\lambda_T = 2c_s^2/(G\Sigma)$ is the most unstable (Toomre) wavelength, with $m_{\text{frag}}/M_{\text{analytical}}$ log-normally distributed around 1.3 with a scatter of 0.3 dex. We evaluate $M_{\text{analytical}}$ following their prescription, with Σ and c_s azimuthally averaged over the radial slice $R_{\text{frag}} \pm R_{\text{eff}}$ at t_{identify} , each

Table 2. The bound-object census

Object	t_{identify}	m_{frag}	R_{frag}	R_{eff}	$\min \alpha_{\text{vir}}$	$m(t_{\text{end}})$	Fate / notes
	(kyr)	(M_J)	(au)	(au)		(M_J)	
F1	2.46	2.36	122	4.4	1.03	8.9	survives, absorbs M1 at 3.53 kyr
F2	2.56	2.62	125	4.0	0.99	19.3	survives, accretes debris
F3	2.70	2.97	117	5.1	1.02	20.0	survives
F4	2.95	2.53	124	4.5	1.03	6.2	survives
F5	3.19	2.77	130	4.0	1.09	3.9	survives
F6	3.24	1.77	156	4.1	1.16	2.4	survives, being scattered at t_{end}
F7	3.24	1.38	134	3.2	1.15	1.7	survives
D1	2.68	3.13	126	4.9	1.28	...	disrupted at 2.87 kyr
M1	3.41	1.59	177	3.9	1.27	...	merges into F1 at 3.53 kyr

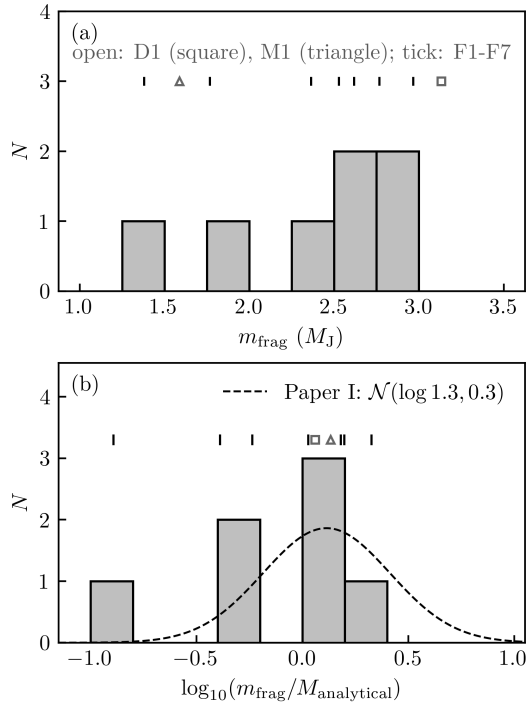


Figure 3. Initial fragment masses. (a) Histogram of m_{frag} for F1–F7, with ticks marking the individual fragments and open symbols marking D1 (square) and M1 (triangle). (b) Initial masses normalized by $M_{\text{analytical}} = \Sigma \cdot \lambda_T \cdot 2c_s / \Omega$ (Equation (40) of Paper I, evaluated with the isothermal sound speed and the slice averages defined in Section 3.2). The dashed curve is the log-normal found by Paper I, scaled to $N = 7$. Ticks mark individual fragments, and open symbols mark D1 and M1, which are not outliers in initial mass.

1.06 and their standard deviation is 0.43 dex. With only seven fragments the comparison is necessarily coarse. A one-sample Kolmogorov–Smirnov test against that log-normal gives $p = 0.44$, so the null hypothesis that the ratios are drawn from it cannot be rejected.

3.3. Accretion

We measure fragment growth by the net rate of change of the bound mass, smoothed over the orbital shear time (Section 2.4). The smoothing is necessary because gas near the boundedness surface of Equation (1) crosses it repeatedly in both directions. The total exchange of cells between a bound object and its surroundings therefore exceeds the net rate by a factor of several without any corresponding transport of gas, and tracing individual cells shows that these fluctuations lose coherence within a few output intervals.

Figure 4a shows the smoothed growth rates. Two measured facts fix the natural rate scale. First, the thermal radius R_B exceeds the Hill radius for every fragment at every epoch, with per-fragment medians of R_B/R_H between 3.5 and 10.5 across the seven fragments, so tides rather than thermal pressure limit what a fragment can hold and growth is Hill-limited. Second, R_H exceeds the local scale height H by up to about a factor of two, so the feeding geometry is two-dimensional and the natural rate scale is $\Sigma_{\text{loc}} \Omega R_H^2$. A single scaling then describes the whole population, with a median efficiency $C_{\text{eff}} \approx 4$ over rates that span more than two decades and per-fragment medians between 1 and 6 (Figure 4b). Growth therefore runs at a few times the nominal Hill rate, fed by the dense gas in which the fragments remain embedded. We emphasize, however, that this is an effective scaling for the smoothed growth of the bound mass, not an instantaneous rate of gas crossing the Hill surface.

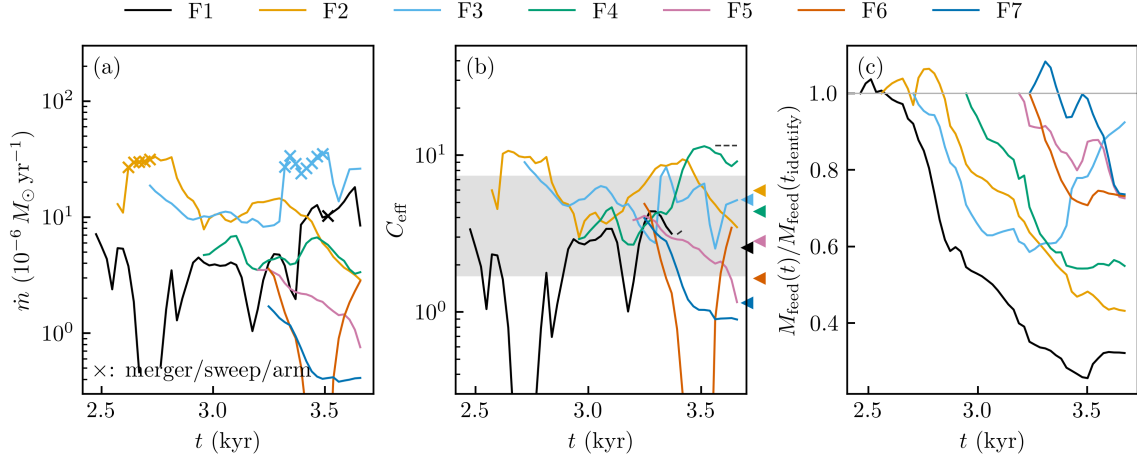


Figure 4. Accretion. (a) Net growth rate of each fragment mass, smoothed over the orbital shear time (Section 2.4) with no intervals masked, so the episodes remain visible. Crosses mark the three episodes for which a mechanism can be identified, namely F2’s debris sweep at 2.6–2.7 kyr, F3’s arm crossing at 3.3–3.5 kyr, and the M1 merger into F1 at 3.5 kyr. A few intervals of negative smoothed rate, three for F1 near 2.7 kyr and two for F6 during its scattering, cannot be shown on the logarithmic scale. (b) The Hill efficiency $C_{\text{eff}} = \dot{m}/(\Sigma_{\text{loc}}\Omega R_{\text{H}}^2)$. The shaded band is the 16th to 84th percentile range of all measurements of all fragments and the left-pointing markers are the per-fragment medians. Intervals affected by the merger or by close encounters between clumps are omitted. For F1 the curve is dashed once its averaging window reaches the merger, from 3.4 kyr onward, because the smoothed rate there measures the binding of gas delivered by M1 rather than accretion from the disk. (c) Ambient mass M_{feed} of each fragment’s feeding annulus (Section 2.4), normalized to its value at the fragment’s identification.

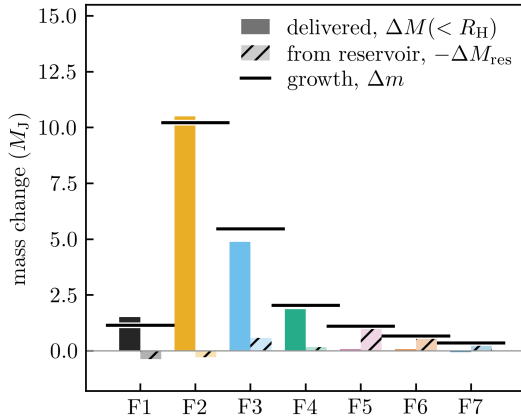


Figure 5. Delivery and binding, measured over the first 434.7 yr after each fragment is identified (Section 2.4). Solid bars show the change of the total gas content of the Hill sphere. Hatched bars show minus the change of its unbound part, which is positive where the fragment drew that reservoir down and negative for F1 and F2, whose unbound content grew. Black lines show the measured growth of the bound mass, which by Equation (9) is the sum of the two bars as plotted, since the hatched bar already carries the minus sign. The early fragments and F4 are fed as they grow, whereas F5–F7 grow almost entirely by consuming the gas already inside their Hill spheres.

The distinction matters because a fragment can bind only gas that is already inside its Hill sphere. Growth therefore has two steps. Gas is first *delivered* into the

Hill sphere from the surrounding disk, and it then becomes *bound* to the fragment. The Hill sphere acts as a reservoir between the two steps, and which step limits the growth depends on whether that reservoir is refilled as fast as it is consumed. Since $M_{\text{res}} = M(< R_{\text{H}}) - m$ by definition, the change of bound mass over any interval satisfies

$$\Delta m = \Delta M(< R_{\text{H}}) - \Delta M_{\text{res}}, \quad (9)$$

which splits the growth into the change of the Hill inventory and the change of the unbound reservoir. We then measure the two terms separately.

Binding is the immediate growth channel for every fragment, since 88 to 100% of the newly bound cells were already inside the Hill sphere at the previous snapshot. What differs across the population is delivery, and Figure 5 shows the two terms of Equation (9) for all seven fragments. Over the first 434.7 yr of each fragment’s life, the Hill spheres of the early fragments F1–F3 gain mass at about the rate the fragments themselves do, between 1.5 and 10.5 M_{J} against growths of 1.1 to 10.2 M_{J} , so the disk refills the reservoir as fast as the fragment empties it and delivery is not the limiting step. The Hill spheres of the three latest fragments gain almost nothing over the same interval, less than 0.1 M_{J} , while their bound masses grow by 0.35 to 1.1 M_{J} . That growth is drawn from the gas already present in their Hill spheres, which declines by the corresponding amount. These fragments are consuming a fixed reservoir rather than being fed.

For F5–F7 this interval is essentially their whole life to the end of the run. F4, though intermediate in formation time, is fed like the early fragments, its Hill sphere gaining mass about as fast as it grows. The binding step itself is the envelope process that Section 3.6 follows.

The contrast in delivery reflects formation environment and history. The late fragments condensed from the arms of a disk that the earlier generation was already draining. Compared at the same age (Section 2.4), the ambient density around them, measured as the median density of ambient gas in a shell between 1.2 and $2.5 R_H$, is five times lower, the local surface density is two times lower, and the total inflow into their Hill spheres is more than an order of magnitude smaller than what the early fragments received. The deficit then feeds back on itself, since less delivery means less mass, a smaller Hill cross-section, and still less access. The late fragments accordingly remain poorly supplied to the end of the run.

Supply also regulates the early fragments, and that supply is local. For F1–F4 we trace each arriving cell back five snapshots before it first crosses the Hill surface and measure its radial offset from the fragment’s orbit in units of the Hill radius, both taken at that earlier time. Half of the arriving gas originates within 1.3 to $1.6 R_H$ of the orbit and 95% within 2.0 to $2.4 R_H$. This measured feeding width is what motivates the $a \pm 2 R_H$ feeding annulus of Section 2.4. These feeding annuli are being exhausted. Every fragment’s annulus drains, ending at 32–92% of its value at identification (Figure 4c). The fragments themselves consume the gas, since by the end of the run the bound objects contain $62 M_J$, or 31% of all the gas in the simulation, and each newly formed fragment is another competitor for the same annuli. In addition, the global gas flows driven by the dynamics of the disk and the fragments together rearrange what remains. This redistribution temporarily refills some annuli while draining others, e.g., F7’s briefly exceeding its initial value. Growth is therefore supply-limited on the timescale of disk evolution, 10^{-1} – 10^0 Myr, and the long-term fate of each fragment is set less by its present accretion rate than by what the disk can still deliver. Section 3.7 makes the quantitative prediction.

Discrete episodes punctuate the growth of the early fragments. F2 accretes $+3.7 M_J$ of dense debris between $t = 2.61$ and 2.73 kyr, left by a transient over-density that had dispersed shortly after the onset of fragmentation. F3 crosses a spiral arm between $t = 3.31$ and 3.50 kyr and gains $+5.4 M_J$. The merger with M1 delivers $+1.9 M_J$ to F1 at $t = 3.5$ kyr. Because the growth rate is averaged over a window of about 270 yr, the merger also affects F1’s measured efficiency for roughly

half a window on either side, which is why the dashed segment in Figure 4b begins near 3.4 kyr rather than at the merger itself.

3.4. Migration

Figure 6a shows the orbital histories. Migration is bidirectional and fast. Three fragments move outward on average (F1, F2 and F4) and four move inward (F3, F5, F6 and F7), at mean rates up to 0.05 au yr^{-1} . For the three latest fragments, followed for roughly a quarter to two-fifths of an orbit, the mean inward drift is comparable to the epicyclic radial motion of their eccentric orbits, so their secular direction is less certain than that of the longer-lived fragments. Compared with the linear Type-I timescale t_I of Section 2.4, only F3 drifts as linear theory expects, inward with $|a/\dot{a}|/t_I = 0.9$. The median ratios of the others lie between 0.1 and 1.6, corresponding to drifts up to an order of magnitude faster than the linear rate or directed outward where linear theory predicts inward. The premises of linear theory are doubly violated here, because the fragments are super-thermal and the disk is neither laminar nor smooth, so we regard the comparison as a reference scale rather than as a test of the theory.

The angular-momentum budget identifies what drives the drift. For every fragment, the net change of total angular momentum consists of the accreted angular momentum of the joining cells, less that carried away by the leaving cells, plus the external torque from the ambient gas and that from the other clumps. Here, we focus on the latter two sources to understand the dynamics driving the orbital drift of fragments. Figure 6c attributes the external torque to its two sources following Equation (4). For five fragments (F1–F4 and F7) the torque from ambient gas dominates, while for F5 and F6 the gravity of the neighboring clumps dominates, so that their orbital evolution is few-body dynamics rather than a disk torque. F6 is the extreme case. Its nearest neighbor throughout is F2, which recedes at first and then closes on it steadily while F6’s eccentricity climbs from its identification at $t = 3.24$ kyr to reach $e = 0.74$ at the end of the run. The scattering is still in progress when the simulation stops.

3.5. Spin

Figure 7 summarizes the rotational state of the fragments. The six fragments untouched by the merger spin up, with transient reversals and obliquity excursions during their accretion episodes that are visible for F2 and F3 in Figure 7a and b. By the end of the run their rotational support spans $\beta_{\text{spin}} = 0.09$ to 0.15 across the six. Their rotation is differential, the rate declining outward with a logarithmic slope q_Ω between -1.09

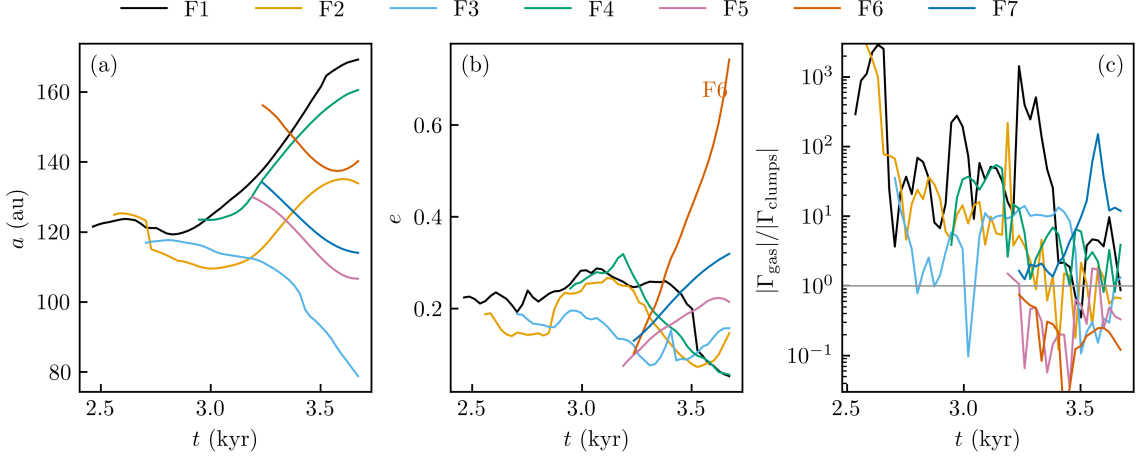


Figure 6. Migration. (a) Orbital radius of each fragment. (b) Orbital eccentricity, with the labeled curve showing F6, which is being scattered when the run ends. (c) Ratio of the gravitational torque from ambient gas to that from the other clumps (Section 2.4), on the bound members of each fragment. The horizontal line marks equality.

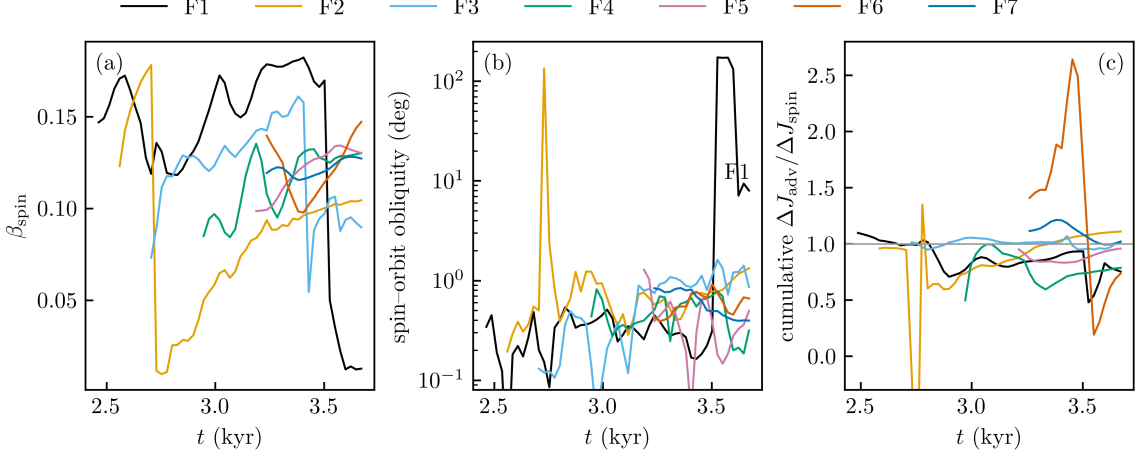


Figure 7. Spin. (a) Rotational support β_{spin} (Section 2.4). (b) Spin-orbit obliquity, on a logarithmic scale, with the post-merger F1 labeled. (c) Running total of the spin angular momentum carried by joining and leaving cells, as a fraction of each fragment's spin change accumulated to the same time (Section 2.4). The horizontal line marks unity.

and -0.12 across the six, and -1.43 for the post-merger F1. In general, these are pressure-supported bodies that rotation flattens only mildly. The rotation is also ordered, in that the mean azimuthal velocity about the spin axis exceeds its dispersion by factors of a few, and it is aligned, with spin-orbit obliquities mostly below 1° as shown in Figure 7b. Accretion supplies this spin. Integrated over each fragment's life, the spin angular momentum carried by the joining cells, less that carried away by the leaving cells, accounts for most of the change of the spin. Figure 7c shows the cumulative total, which fluctuates more widely at earlier times because the accumulated change of the spin in its denominator is then still small.

F1 is the exception. The merger leaves it slowly rotating, with $\beta_{\text{spin}} = 0.013$, weakly ordered, and tilted, its spin axis inclined by 8° to its orbital axis, the only final obliquity above 1.5° in the sample. A single merger, here with a mass ratio of 0.29 between M1 and F1 just before they combine, resets the spin state that accretion had built.

3.6. Interior structure

Figure 8 shows the density, temperature and entropy profiles of every fragment at every analyzed epoch. The seven objects share one structure and one evolutionary trend. Each is a centrally condensed body a few astronomical units across surrounded by an extended envelope. Over the life of each fragment the central density

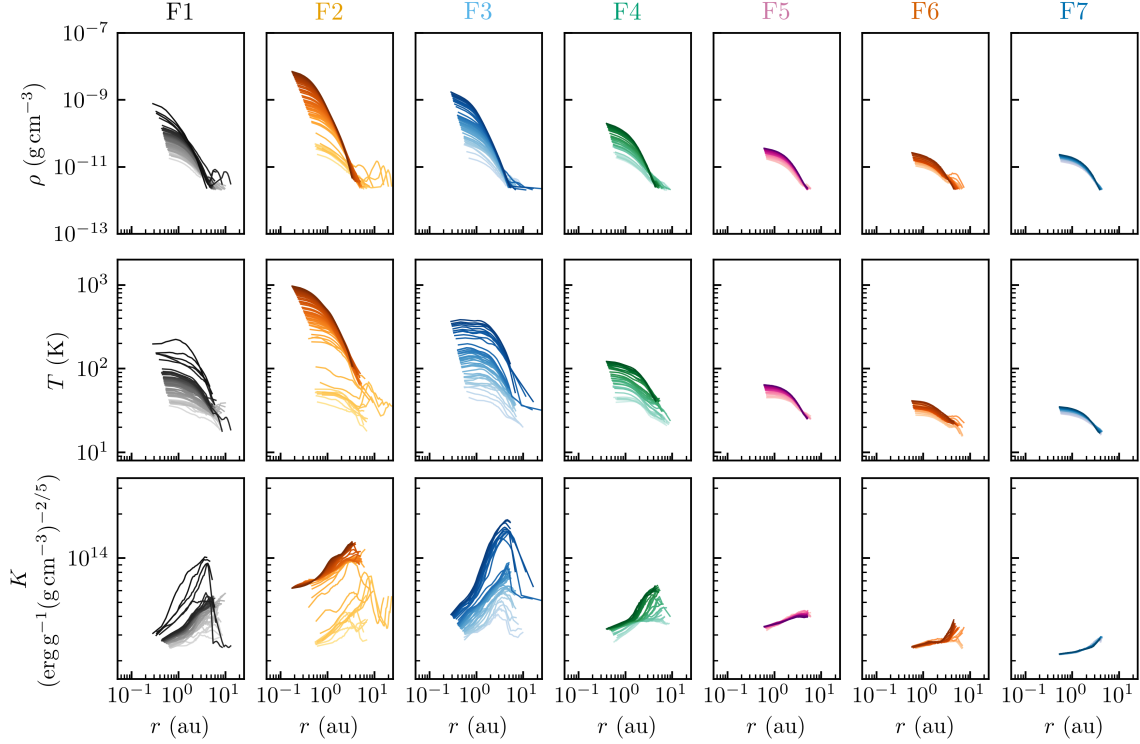


Figure 8. Interior profiles of all seven fragments at every analyzed epoch: density (top row), temperature (middle), and adiabat label $K = (\gamma - 1)u\rho^{1-\gamma}$ (bottom, in units of $\text{erg g}^{-1}(\text{g cm}^{-3})^{-2/5}$), as shell medians about each fragment’s density peak (Section 2.4). Within each column the line color runs from light (identification) to dark (end of the run) in a per-fragment color sequence. In the bottom row the final, monotonic profiles of F1 and F3 are partly overlain by the immediately preceding post-event epochs, drawn at nearly the same dark color.

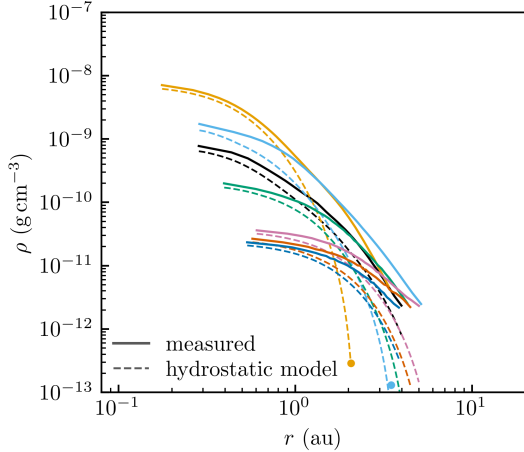


Figure 9. Measured density profiles of all seven fragments at the end of the run (solid), compared with hydrostatic models integrated from each fragment’s measured $K(m_r)$ and central density (dashed). The dashed curves are clipped to each fragment’s measured radial range, and a dot marks the radius where a model reaches zero pressure, drawn only for F2 and F3, whose surfaces fall inside this range. Colors are as in Figure 2.

temperature climbs from 28–44 K at identification to 35–990 K at the end of the run. The strongest accretors heat the most, F2 reaching 990 K, whereas the late fragments barely heat at all. The adiabat label K in general rises outward through the inner half of each fragment’s mass at every epoch, so the fragments are entropy-stratified from birth, and they become progressively more stable against convection as they grow, most of each interior ending Schwarzschild-stable. We further confirm that thermal energy dominates the support throughout: the ratio $U/|E_P|$ declines from between 0.72 and 0.87 at identification to between 0.63 and 0.74 at the end of the run.

If a fragment’s interior is a hydrostatic structure supported by the pressure of gas at its measured entropy, then integrating the equation of hydrostatic equilibrium in the interior mass coordinate m_r , which is the bound mass enclosed within radius r ,

$$\frac{dP}{dm_r} = -\frac{Gm_r}{4\pi r^4}, \quad \frac{dr}{dm_r} = \frac{1}{4\pi r^2 \rho}, \quad \rho = \left[\frac{P}{K(m_r)} \right]^{1/\gamma}, \quad (10)$$

with each fragment’s own measured $K(m_r)$ and central density, must reproduce its measured density pro-

729 rises by up to two and a half decades, while the central

file. Figure 9 shows that the models generally do. The innermost mass is closest to thermal hydrostatic balance, since ordered rotation, measured as the shell average of $\langle v_\phi \rangle^2 r / (Gm_r)$ over the inner half of the mass, supplies only about a tenth to a quarter of the support there, while in the envelope, rotation and the unsettled motion of newly arrived gas contribute support that a purely thermal model omits. Taken together, the approximately matched inner region and the underestimated outer part suggest a centrally condensed, entropy-stratified interior in thermal hydrostatic balance, surrounded by an accreting, rotating envelope that is not.

The hierarchy of the three timescales defined in Section 2.4 demonstrates the physics here. Across the seven fragments, $t_{\text{ff}} \sim 10^0\text{--}10^1$ yr, $t_{\text{acc}} \sim 10^3$ yr, and $t_{\text{KH}} \sim$ a few $\times 10^4$ yr. Because $t_{\text{ff}} \ll t_{\text{acc}}$, a fragment stays in hydrostatic balance while it grows. Because $t_{\text{acc}} \ll t_{\text{KH}}$, it grows far faster than it can cool, so its structure is mostly shaped by accretion, and the compression that accretion drives is nearly adiabatic: a parcel of gas largely keeps the entropy it had when it was bound, since convection is suppressed by the stratification itself and the opaque interior lets out only a few percent of the accretion power. The entropy profiles in the bottom row of Figure 8 can then be read as a record. In space, the outward rise of K traces a stack in which the deepest layers are the earliest, lowest-entropy arrivals and each new layer settles on top with slightly higher entropy, so the amplitude of the stratification implies how much the fragment has accreted. The outermost parts of the fragments develop transient entropy-decreasing layers, where gas assembly is entangled with the complicated dynamics of the surrounding disk. The deepest such layers appear during and just after a delivery event, namely F1’s merger, F2’s debris sweep and F3’s arm crossing, with shallower inversions in F3’s first epochs while its initial envelope settles. In time, the interior curves of the starved late fragments lie on top of one another from epoch to epoch, frozen because little is being added, while the profiles of the strongly fed early fragments extend outward and rise at their outer ends as accretion stacks new layers, severalfold for F2 and F3. Only the interiors of the strongest accretors drift slowly upward, the residue of irreversible heating during their most rapid growth.

Two consequences follow for the later evolution of these objects. The first concerns their thermal state, and it bears on the distinction between the hot-start and cold-start initial conditions used in models of giant-planet evolution, which differ in how much of the entropy of the accreted gas a young object retains (e.g.,

A. Burrows et al. 1997; M. S. Marley et al. 2007; D. S. Spiegel & A. Burrows 2012; G.-D. Marleau & A. Cumming 2014). That distinction is observable, because a hot start leaves an object markedly more luminous at a given mass and age. In the final profiles entropy rises outward with the adiabat, so a fragment’s central value is the lowest it holds. In our simulation, the specific entropies at the center at the end of the run are $s = 12.0, 12.9, 12.3, 12.2, 12.2, 11.9$ and $11.8 k_{\text{B}}$ per baryon for F1 to F7. All lie well above the 9.2 to $10.5 k_{\text{B}}$ per baryon that those models require of the directly imaged planets (G.-D. Marleau & A. Cumming 2014): values that imply hot-start initial conditions should these objects eventually become gas giants. We will discuss their fates in Section 3.7.

The second concerns survival. The surviving interiors are robust against the stellar tide. By the end of the run the mean density inside each fragment’s half-mass radius exceeds the local tidal density $3M_*/(2\pi a^3)$ by a factor between 60 and 1.5×10^4 . The cores built by this quasi-adiabatic compression are therefore effectively immune to destruction by the stellar tide at their present orbits. Survival is decided by the dynamical environment, as Section 3.1 showed.

3.7. Conditional final masses

The final masses of these objects are only partly decided within the simulated span: F2 and F3 exceed the deuterium-burning threshold ($13 M_{\text{J}}$), and the gas budget that constrains any future growth is limited. The simulation holds $202 M_{\text{J}}$ of gas, of which $62 M_{\text{J}}$ is already in the seven fragments and $140 M_{\text{J}}$ is ambient ($88 M_{\text{J}}$ of it between 60 and 200 au), with only $8.6 M_{\text{J}}$ of ambient gas currently inside the seven Hill spheres (each cell assigned to the fragment for which its distance in units of that fragment’s Hill radius is smallest, so overlaps are not double counted). The run does not, however, directly give a unique final mass for each fragment, since the simulation duration is short compared to the disk lifetime of $10^{-1}\text{--}10^0$ Myr. We therefore report *conditional* survivor masses. We integrate the growth law forward for every fragment at once, drawing on a single shared reservoir,

$$\frac{dm_i(t)}{dt} = C_i \Sigma_i(t) \Omega_i(t) R_{\text{H},i}^2(t), \quad (11)$$

$$\frac{dM_{\text{pool}}(t)}{dt} = - \sum_i \frac{dm_i(t)}{dt}, \quad (12)$$

where C_i is an assumed constant efficiency of order unity, the scale set by the measured values of Section 3.3, and M_{pool} is the accessible reservoir, meaning the part

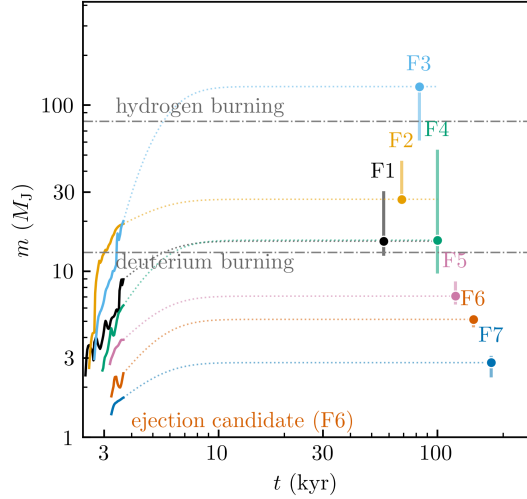


Figure 10. Final masses. Solid curves are the measured mass histories, on a logarithmic time axis, and dotted curves continue each fragment along its computed mass evolution, which saturates once the shared reservoir is consumed. Filled circles mark the conditional masses at 0.1 Myr from the calculation of Section 3.7. The thin bars span the adopted value together with the variants in which the lifetime-median surroundings, the lifetime-median efficiencies, or F1’s literal end-of-run efficiency separately replace the adopted end-state inputs. These are assumption sensitivities, not confidence intervals, and the markers are offset horizontally for legibility. The labeled annotation marks F6, whose ongoing scattering may end in ejection. Dash-dotted lines mark the deuterium-burning threshold and the hydrogen-burning boundary.

of the ambient gas that the disk can deliver to the fragments. The Hill radius follows the growing mass while the orbital frequency follows the orbit,

$$R_{H,i}(t) = a_i(t) \left[\frac{m_i(t)}{3M_\star} \right]^{1/3}, \quad \Omega_i(t) = \left[\frac{GM_\star}{a_i(t)^3} \right]^{1/2}, \quad (13)$$

and the surface density available to each fragment declines as the disk is drained,

$$\Sigma_i(t) = \Sigma_i(0) M_{\text{pool}}(t)/M_{\text{pool}}(0). \quad (14)$$

This closure is ad hoc: it ties the surface density around each fragment to the one reservoir they share, standing in for delivery physics that a model this simple cannot derive from first principles. And because the disk gas need not all end up in the fragments, the reservoir may be only a fraction of what the disk holds: we set $M_{\text{pool}}(0) = f_{\text{acc}} M_{\text{amb}}$, with $M_{\text{amb}} = 140 M_J$ the measured ambient mass and f_{acc} the accessible fraction.

Three inputs remain undetermined: the efficiencies C_i , the orbits $a_i(t)$, and the accessible fraction f_{acc} . For simplicity, we fix each orbit at its end-of-run radius, and

set $f_{\text{acc}} = 1$, i.e., assuming the whole current gas budget is available to the fragments and no new gas arrives. We set each C_i to the median of the fragment’s measured efficiency over its last five clean intervals: $C_i = 4.1, 4.1, 4.8, 9.1, 1.9, 1.8$ and 0.9 for F1 to F7, where for F1 we use the last intervals unaffected by the merger, since its literal end-of-run value (11.5) measures the binding of M1’s debris rather than accretion from the disk (Section 3.3). For the starved F5–F7 these efficiencies describe the binding of gas already resident in their Hill spheres rather than fresh delivery, so the model, if anything, overstates their draw on the shared pool. Each integration starts from the measured end state of every fragment, namely its bound mass $m_i(0)$, its orbital radius $a_i(0)$, and a starting surface density $\Sigma_i(0)$ set to the median of its local Σ_{loc} over the last five snapshots, and advances to 0.1 Myr with a fourth-order Runge–Kutta scheme in 2500 steps, a resolution at which the final masses change by less than 0.01% if the number of steps is doubled.

Figure 10 shows the outcome, and it is a division of the entire ambient budget: because the supply falls only as the shared pool drains, accretion stops only when the pool is gone, and here it is gone quickly, 99% consumed within 7.2 kyr. Each fragment’s share follows its initial rate $C_i \Sigma_i(0) \Omega_i R_{H,i}^2(0)$, the Hill factor favoring the already massive. F3, which combines the densest end-of-run surroundings of the seven with the largest bound mass, takes 78% of the ambient mass and reaches $130 M_J$, crossing the hydrogen-burning boundary ($\approx 80 M_J$) to become a low-mass stellar companion under these assumptions. F2 settles at $27 M_J$ and remains a brown dwarf. F1 and F4 end just above the deuterium threshold, at $15 M_J$ each, and the late fragments F5–F7 remain gas giants at $7.1, 5.1$ and $2.8 M_J$. These masses are conditional on the stated assumptions, and their firmness differs: separately substituting lifetime medians for the end-state surroundings and for the efficiencies moves individual masses by factors of a few (F3 between 61 and $133 M_J$, F4 between 10 and $54 M_J$). In every variant F2 remains a brown dwarf, F5–F7 remain giants, and F3 ends as the most massive object, although in the lifetime-median-surroundings variant F4’s gain slightly exceeds F3’s, and the deuterium classification of F1 and F4 is not robust, their variants straddling the threshold. The two caveats that follow both bear mainly on F3.

The first caveat is gap opening, and for the dominant consumer its effect runs in one direction: a gap around F3 would lower F3’s mass, whether the withheld gas is redistributed to the other fragments through the shared pool or never delivered at all. Every frag-

ment already satisfies the thermal condition for opening a gap, with Hill radii exceeding the local scale height by factors of 1.0 to 2.1 at the end of the run, so in a low-viscosity laminar disk objects this massive would clear their co-orbital gas and throttle their own supply (A. Crida et al. 2006). Two measured properties of this disk resist clean gaps: the torques on the fragments come from spiral structure and from the neighboring clumps rather than from a smooth viscous background (Figure 6c), and the orbits are eccentric, with end-of-run apocenter-to-pericenter spans of 0.8 to 62 Hill radii that, for all but the nearly circular F1 and F4, sweep the co-orbital region rather than resting in it. Consistently, no fragment opens a cleared channel within the simulated 1.2 kyr, and the feeding annuli drain smoothly instead (Section 3.3). Over 10^5 yr, however, even partial gap opening would hold the local surface density below the pool-proportional closure of Equation (14), so the calculation overestimates the gapped fragment’s late accretion: F3’s crossing of the hydrogen-burning boundary is the claim most likely to be undone by a gap.

The second caveat is the fixed orbits. The explicit orbital dependence of the growth rate is weak, since $\Sigma \Omega R_H^2 \propto \Sigma a^{1/2}$ at fixed mass. What migration changes is the surroundings a fragment finds, which the pool-proportional closure ignores. Sustained at its mean rate, F3’s inward drift would carry it into the inner disk, while the sustained outward migrators F1 and F4 climb toward the thinning outer edge. The sign of the net effect on F3 is therefore uncertain, unlike that of gap opening: continued inward drift through dense gas could sustain its dominance, while the scattering and eccentricity pumping that accompany the crowded late disk could equally strand it. The drifts themselves are episode-driven and bidirectional, F2 having reversed direction during the run, so extrapolating any of them over 10^5 yr is unrealistic.

The calculation above deliberately omits mergers, disruption, scattering, and new fragmentation. F6 carries the dynamical channel explicitly: it is being scattered ($e = 0.74$ and rising when the run ends), and ejection would deliver a free-floating object of planetary mass ($\sim 3\text{--}4 M_J$ with its resident envelope). A single fragmenting disk therefore produces multiple distinct outcomes, namely gas giants, brown dwarfs and, if the scattering of F6 ends in ejection, free-floating planets, with a low-mass stellar companion as a fourth and conditional outcome that our calculation reaches only if F3 keeps its full supply, opening no gap and holding its orbit.

4. DISCUSSION

4.1. Comparison with Paper I

The most direct point of contact with Paper I is the initial fragment mass. The two simulations differ in nearly every ingredient that could plausibly set that mass, namely the equation of state, the opacity law, the disk extent and the surface density profile, yet the normalized distribution $m_{\text{frag}}/M_{\text{analytical}}$ is consistent with the Paper I log-normal (Section 3.2). This consistency across setups supports the interpretation of Paper I that the fragmentation scale is set locally, by the thermal state of the collapsing material, rather than by the global disk configuration. The fates, too, repeat the Paper I phenomenology: surviving fragments descend in α_{vir} while dissolving clumps turn around before reaching equilibrium (Figure 2b), provided that the virial parameter is measured on the scale consistent with the equation of state, Equation (2).

The evolution measured here also sharpens the mass question that Paper I left open. Fragments are uniformly born planetary, but growth does not stop at planetary masses: by the end of the run the two most massive fragments already exceed the deuterium-burning threshold of $\approx 13 M_J$ (D. S. Spiegel et al. 2011), and the conditional survivor masses of Section 3.7 spread the population from gas giants to a possible low-mass stellar companion, the latter only if the dominant accretor keeps its full supply. Whether a GI fragment becomes a planet, a brown dwarf, or a low-mass stellar companion is decided not at fragmentation but by how much of the remaining disk gas is ultimately delivered to its Hill sphere.

4.2. Implications for the evolution of GI fragments

Our measurements bear on the accretion, migration, spin, and contraction that models of fragment evolution have to assume. A single Hill-regulated law organizes the growth of the whole population, with $C_{\text{eff}} \approx 4$ and a scatter of a factor of a few, but the growth it predicts is set by how fast the disk delivers gas. That coefficient is the one Z. Zhu et al. (2012) obtained in two dimensions, $\dot{m} = 4 \Sigma \Omega R_H^2$ (Section 1). We reach it independently, in three dimensions and with radiation transport, which suggests the Hill-rate normalization is robust to both. Fragments that form early, in dense surroundings, grow as fast as their Hill spheres are refilled, and their growth is punctuated by discrete events such as a merger, an episode of debris accretion, or an arm crossing. The latest-forming fragments receive more than an order of magnitude less gas and grow instead by binding the gas already inside their Hill spheres. Accretion prescriptions that ignore this dependence of supply on

environment and formation time will predict the growth of both groups incorrectly.

Migration is neither uniformly inward nor Type-I-like. Three of seven fragments migrate outward on average, the rates exceed the linear estimate by up to an order of magnitude, and for two of the three late-forming fragments (F5 and F6) the dominant driver is the gravity of the other clumps rather than the disk, a few-body process that prescriptions based on smooth disk torques cannot represent. This is where we depart from Z. Zhu et al. (2012), who found uniformly inward migration and lost four of thirteen clumps to tidal disruption. Their sample is the marginally fragmenting one of one or two clumps per disk, which they note does not suffer the strong clump-clump scattering of multiple fragmentation, a regime they set aside and our disk populates. Their disruptions also happen at the small radii their clumps migrate to, whereas our fragments stay between 79 and 169 au, where the stellar tide is weak (Section 3.6). Three-dimensional simulations that follow several fragments at once likewise find the migration stochastic in direction (N. Kubli et al. 2026). Rapid inward delivery, a key ingredient of tidal downsizing, occurs here for only part of the population, while F6 illustrates the opposite outcome, in which scattering between clumps may end in ejection rather than orderly inward migration (Section 3.4).

Competition between fragments also sets the masses these objects reach. A clump with sole use of its annulus reaches the isolation mass of order $0.1 M_{\odot}$ that motivates the brown-dwarf expectation of Section 1. Seven fragments drawing on one reservoir divide it instead: in Section 3.7 the same growth law gives the isolation-scale mass to the dominant consumer alone, while the late-forming fragments, which arrive to find their annuli already draining, stay at planetary masses. Whether this channel makes planets or brown dwarfs is therefore decided by how many fragments share the disk, not by the growth law itself.

The interiors, finally, are not the non-rotating, purely thermally supported polytropes commonly assumed as initial conditions for contraction calculations. Idealized models of fragment interiors usually distinguish a thermally supported and slowly rotating structure from a rotationally flattened one (e.g., A. C. Boley et al. 2010; M. Galvagni et al. 2012). Our fragments combine the two, with a thermally supported, stratified and convectively stable core beneath a rotating accretion envelope. Only the merger-processed F1 resembles the purely thermal case. While the fragments remain embedded their interiors cannot cool, because radiative transport falls far short of the accretion power (Section 3.6). Contraction

calculations for GI-born objects should therefore start from high-entropy, rotating, stratified states, and their subsequent evolution will have to shed or reorganize the angular momentum stored in the envelopes.

One extrapolation follows from the spin measurements. We emphasize that it is a prediction that this run cannot test. The six unmerged fragments end the run rotating at 40–60% of the breakup rate at their half-mass radii, meaning $\Omega_{\text{spin}}(R_{\text{half}}) [G(m/2)/R_{\text{half}}^3]^{-1/2} \approx 0.4$ – 0.6 , and are still spinning up by accretion and contraction. If their specific angular momenta are conserved during further contraction, rotation becomes limiting at radii of order 0.1–1 au, where the natural outcome is the shedding of material into a circumplanetary disk (e.g., A. C. Boley et al. 2010; M. Galvagni et al. 2012). On this extrapolation, circumplanetary disks would be a generic by-product of the contraction of GI fragments, except where mergers intervene. The single merger in our sample resets the remnant’s spin and tilts it (Section 3.5), so merger-processed fragments would begin contraction with little rotation and misaligned spin axes. Testing this prediction requires following fragments through their second collapse, which the present run does not reach.

4.3. *Observational signatures for planetary-mass free-floating objects*

Our run is too short to decide F6’s fate. Its eccentricity climbs steadily to $e = 0.74$ by the end of the run, driven by its nearest neighbor rather than by the disk (Section 3.4), but the encounter is unfinished, and whether it ends in ejection or in disruption by its neighbor’s tide (A. Čalović et al. 2026) depends on the later dynamical evolution of the whole system, including the inner disk that this work does not model. Should it end in ejection, what leaves is the free-floating planetary-mass object Sections 3.5 and 3.6 measured.

Such an object would be distinctive. It leaves while still pre-collapse, so this channel should place objects that are over-luminous for their mass in the youngest star-forming regions, still carrying their rotating envelopes. Isolated planetary-mass objects are indeed observed with dusty disks (A. Bayo et al. 2017; B. Damian et al. 2025), and one has been caught in an accretion burst (V. Almendros-Abad et al. 2025). A deep survey of Upper Scorpius and Ophiuchus finds of order a hundred candidate free-floating objects between 4 and $13 M_{\text{J}}$ (N. Miret-Roig et al. 2022), and microlensing constrains the free-floating mass function down to Earth masses (T. Sumi et al. 2023), with the caveat that a microlensing event cannot distinguish an unbound object from one on a wide orbit (J. C. Yee & S. J. Kenyon 2025). Our

simulation shows that disk fragmentation is a plausible channel for producing them.

4.4. Caveats

There are a few numerical caveats we must acknowledge. (i) All results derive from a single realization of a single disk, so the population statements rest on seven objects and carry no ensemble variance. (ii) The output cadence (24.15 yr) is coarser than the initial growth time of a young clump, so m_{frag} is biased high by up to one cadence of growth. (iii) The run covers 3.67 kyr, and every evolutionary statement is censored at that horizon, since the reservoirs are still draining, F6’s scattering is unresolved, and the approach of the youngest fragments to virial equilibrium is incomplete.

Besides, the following caveats concern the physical model here. (i) The run omits stellar irradiation and magnetic fields. [Paper I](#) discusses why irradiation is subdominant for the fragmenting outer disk, and non-ideal magnetohydrodynamics is known to alter both the gravito-turbulent state and clump accretion (e.g., [H. Deng et al. 2021](#); [N. Kubli et al. 2023](#)). (ii) The fixed $\gamma = 1.4$ equation of state excludes H_2 dissociation and the second collapse it triggers. This remains consistent through the end of the run (central temperatures stay at or below ≈ 1000 K, although at that temperature vibrational excitation of H_2 already begins to soften γ) but sets the same horizon as the duration: the final contraction of the fragments is beyond this paper. (iii) The simulated disk is also a closed box: nothing replenishes it from an envelope or an outer disk, and the conditional masses of Section 3.7 divide a fixed budget accordingly.

5. CONCLUSIONS

We have followed every bound object in a global radiation hydrodynamic simulation of a fragmenting protoplanetary disk, namely seven surviving fragments, one disrupted clump and one merged clump, over the final ≈ 1.2 kyr of the simulation, closing mass and angular-momentum budgets per object. Our main conclusions are:

- Fragments form at $1.4\text{--}3.0 M_{\text{J}}$, and their masses normalized by the local analytic estimate ($M_{\text{analytical}} = \Sigma \cdot \lambda_{\text{T}} \cdot 2c_{\text{s}}/\Omega$) are consistent with the log-normal distribution of [Paper I](#) despite a different equation of state, opacity, disk model and fragmentation environment. The fragmentation mass scale is consistent with being set locally, independent of the disk model. Survival is not decided by initial mass, since the disrupted and merged clumps are not outliers in initial mass.

- Growth proceeds in two steps, the delivery of disk gas into each Hill sphere and the binding of gas already inside it, the latter being the immediate growth channel for every fragment. A single law, $\dot{m} \approx 4 \Sigma_{\text{loc}} \Omega R_{\text{H}}^2$ with a scatter of a factor of a few, describes all seven fragments. Growth is limited by tides rather than by pressure ($R_{\text{B}} > R_{\text{H}}$), its geometry is two-dimensional ($R_{\text{H}} > H$), it is regulated by feeding zones extending $\approx 2 R_{\text{H}}$ on each side of the orbit, and it is punctuated by three episodes for which a mechanism can be identified: a merger, a debris sweep and an arm crossing. The spread across the population is set by delivery. The latest-forming fragments, born in surroundings that the earlier generation had already drained, receive more than an order of magnitude less gas at the same age and grow by drawing down the reservoirs already inside their Hill spheres.
- Migration is bidirectional and fast, and only one of seven fragments drifts at the Type-I rate and in the Type-I direction. Angular momentum is exchanged with the disk chiefly through the accreted gas itself, which carries almost all of the change of the total angular momentum for every fragment but F6. The remaining torque comes from the ambient gas for five fragments and, for F5 and F6, from the gravity of neighboring clumps. One fragment is being scattered to $e = 0.74$ when the run ends.
- Accretion sets the spin. The six unmerged fragments spin up to $\beta_{\text{spin}} = 0.09\text{--}0.15$, rotate differentially with rates declining outward, and have spin axes aligned with their orbits to within 1.5° . The gas that joins each fragment accounts for most of its spin change. The single merger in the sample resets the spin of the remnant and tilts it.
- The interiors are entropy-stratified and largely convectively stable, and cool slowly. Their luminosities are a few percent of the accretion power. They grow one to two orders of magnitude faster than they could contract by cooling, so the compression that builds them is driven by the mass they add rather than by the energy they lose, and the work of that compression is what heats them. Hydrostatic models built from the measured entropy profiles generally track the inner parts of the fragments with no free parameters, while the accreting envelopes require support beyond gas pressure. Such a scenario favors hot-start initial conditions for the survivors that become gas giants. The quasi-adiabatic compression of the survivors

leaves cores that are robust against the stellar tide at their present orbits.

- Final masses are conditional, and we report them as such. Integrating the growth law forward against the measured gas budget (202 M_J in total, of which 140 M_J is ambient), with the supply tied to a shared reservoir, the measured end-state efficiencies and surroundings, fixed orbits and the whole ambient budget accessible, consumes the reservoir within 10^4 yr and divides it among the fragments according to their initial accretion rates: F1–F7 reach 15, 27, 130, 15, 7.1, 5.1 and 2.8 M_J at 0.1 Myr. F2 is a brown dwarf, and the late fragments F5–F7 remain giant planets in every variant we ran. F1 and F4 land on the deuterium-burning threshold, their variants straddling it, and F3, which ends as the most massive object in every variant, crosses the hydrogen-burning boundary under these assumptions. F3’s outcome carries

the two caveats: gap opening, which this super-thermal population would attempt and which can only lower F3’s mass, and its fast inward drift, whose net effect on that mass is of uncertain sign. F6, which is being scattered, may be ejected and would then join the free-floating planet population at its present 3 to 4 M_J , resident envelope included.

The natural next steps are simulations that reach the second collapse of the fragments, including H_2 dissociation, longer durations, and the magnetic fields and stellar irradiation omitted here. Such calculations would test the prediction that contraction converts the measured spins into circumplanetary disks, and would determine where the termination of accretion leaves the final mass function.

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